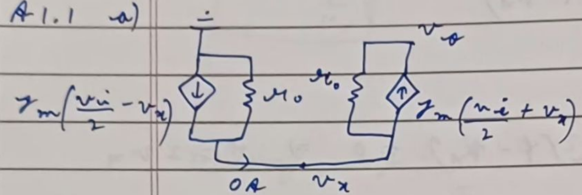
Name	Amogh G. Okade
Roll number	EP22B020

1.1:

24.1.26

Assignment - 1

A1.1 a)

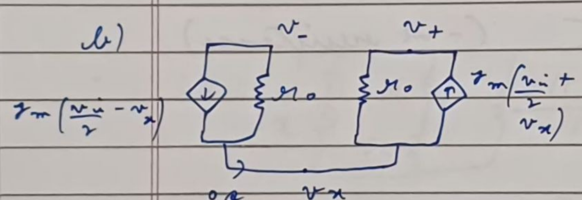


$$\frac{v_x - 0}{r_o} = g_m\left(\frac{v_i}{2} - v_x\right)$$

$$\frac{v_o - v_x}{r_o} = g_m\left(\frac{v_i}{2} + v_x\right)$$

$$\therefore\ v_o = \frac{g_m}{g_{ds}} v_i = g_m r_o v_i$$

b)



$$\frac{v_x - v_-}{r_o} = g_m\left(\frac{v_i}{2} - v_x\right)$$

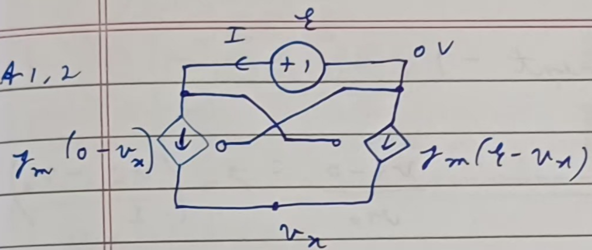
$$g_m\left(\frac{v_i}{2} + v_x\right) = \frac{v_+ - v_x}{r_o}$$

$$\therefore\ v_o = v_+ - v_- = g_m r_o v_i = \frac{g_m}{g_{ds}} v_i$$

- The absence of the 2nd $I_o/2$ current source does not affect the quiscent circuit, since according to KCL, $I_o/2$ is still forced in the left branch.
- In the incremental circuit, b) has no explicit GND. So, we can assume v_- as 0, which gives the same incremental circuit as a). (Except the possibility of current flowing through the GND in a), which doesn't happen here).

1.2:

A1,2



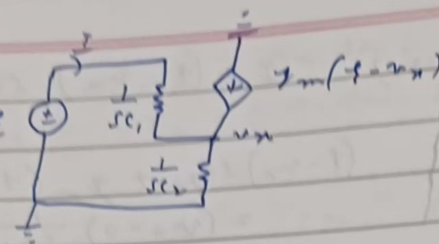
- $g_m(-v_x) + g_m(\xi - v_x) = 0 \Rightarrow \xi = 2v_x$

- $I = g_m(-v_x) = g_m(v_x - \xi) \Rightarrow \xi = -\frac{2}{g_m} I$

- $\therefore \; z_d = -\frac{2}{g_m}$ (-ve resistance)

1.3:

A 1.3



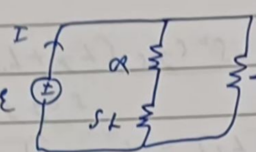
$$I = (\xi - v_x) \times \left(\frac{1}{sC_1}\right)^{-1}$$

$$(\xi - v_x)(sC_1) + g_m(\xi - v_x) = (v_x - 0)(sC_2)$$

solving, $$Z_g = \frac{\xi}{I} = \frac{s(C_1 + C_2) + g_m}{s^2 C_1 C_2}$$

$$Z_g(j\omega) = \frac{-g_m - j\omega(C_1 + C_2)}{\omega^2 C_1 C_2}$$

Z_t :-



$$\frac{s(C_1+C_2)+g_m}{s^2 C_1 C_2}$$

$$\therefore\ Z_t = [\alpha + sL] \,||\, \left[\frac{s(C_1+C_2)+g_m}{s^2 C_1 C_2}\right]$$

$$Z_t = \frac{s^2 L(C_1+C_2) + s[L g_m + \alpha(C_1+C_2)] + \alpha g_m}{s^3 L C_1 C_2 + s^2 \alpha C_1 C_2 + s(C_1+C_2) + g_m}$$

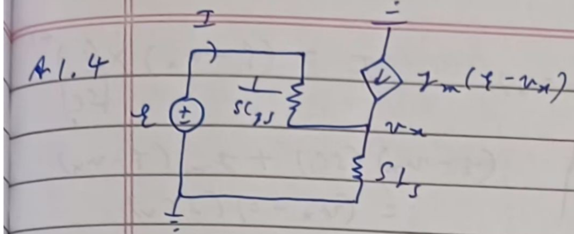
$$Z_t(j\omega_0) = \frac{\alpha g_m - \omega_0^2 L(C_1+C_2) + j\omega_0[L g_m + \alpha(C_1+C_2)]}{g_m - \omega_0^2 \alpha C_1 C_2 + j[\omega_0(C_1+C_2) - \omega_0^3 L C_1 C_2]}$$

$$Z_t(j\omega)_0 = \infty \Rightarrow g_m - \omega_0^2 \alpha C_1 C_2 = 0 = \omega_0(C_1+C_2) - \omega_0^3 L C_1 C_2$$

$$\therefore\ \alpha = \frac{g_m}{\omega_0^2 C_1 C_2}, \quad L = \frac{C_1 + C_2}{\omega_0^2 C_1 C_2}$$

1.4:

A1.4



$$I = (v_g - v_x)\, s C_{gs}$$

$$(v_g - v_x)\, s C_{gs} + g_m (v_g - v_x) = (v_x - 0)\frac{1}{s L_s}$$

Solving, $z_g = \dfrac{v_g}{I} = \dfrac{s^2 C_{gs} L_s + s L_s g_m + 1}{s C_{gs}}$

$$z_g(j\omega) = \frac{1 - \omega^2 C_{gs} L_s + j\omega L_s g_m}{j\omega C_{gs}}$$

$$z_i = s L_g + z_g(s) = \frac{s^2 C_{gs}(L_g + L_s) + s L_s g_m + 1}{s C_{gs}}$$

$$z_i(j\omega) = j\left[\omega(L_g + L_s) - \frac{1}{\omega C_{gs}}\right] + \frac{L_s g_m}{C_{gs}}$$

$$\mathrm{Im}\{z_i(j\omega)\} = 0 \Rightarrow z_i(j\omega) = \frac{L_s g_m}{C_{gs}}$$

and $L_g = \dfrac{1}{\omega_0^2 C_{gs}} - L_s$

1.5:

classmate

Date

Page

A 1.5

$M_1 :- \; I_0 = \frac{1}{2} \mu_n C_{ox n} \frac{W}{L} (V_{GN} - V_{G3} - V_{Tn})^2$

$M_3 :- \; n I_0 = \frac{1}{2} \mu_n C_{ox n} \times \frac{nW}{L} (V_{G3} - 0 - V_{Tn})^2$

$\therefore V_{G3} = V_{Tn} + \sqrt{\frac{2 I_0}{\mu_n C_{ox n} W/L}}$

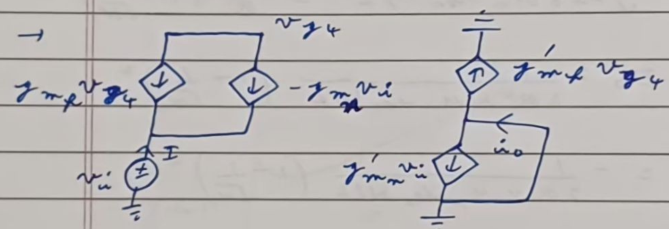
$V_{GN} = V_{G3} + V_{Tn} + \sqrt{\frac{2 I_0}{\mu_n C_{ox n} W/L}} = 2 V_{Tn} + 2\sqrt{\frac{2 I_0}{\mu_n C_{ox n} W/L}}$

$M_4 :- n I_0 = \frac{1}{2} \mu_p C_{ox p} \, n\frac{W}{L} (V_{DD} - V_{G4} - V_{Tp})^2$

$M_2 :- \; I_0 = \frac{1}{2} \mu_p C_{ox p} \frac{W}{L} (V_{G4} - V_{GP} - V_{Tp})^2$

$\therefore V_{GP} = V_{DD} - 2 V_{Tp} - 2\sqrt{\frac{2 I_0}{\mu_p C_{ox p} W/L}}$

→



$I = 0$

$\therefore R_i = \frac{v}{I}$

$\therefore R_i = \infty$

→ $g_{mp} v_{g4} = g_{mn} v_i \Rightarrow v_{g4} = \frac{g_{mn}}{g_{mp}} v_i$

$i_o = g'_{mp} v_{g4} + g'_{mn} v_i = v_i \left[g'_{mn} + \frac{g'_{mp} g_{mn}}{g_{mp}} \right]$

→ $g_{mp} = \sqrt{2 \mu_p C_{ox p} \frac{W}{L} I_0}$; $g'_{mp} = n g_{mp}$

$g_{mn} = \sqrt{2 \mu_n C_{ox n} \frac{W}{L} I_0}$; $g'_{mn} = n g_{mn}$

→ $\therefore \frac{i_o}{v_i} = 2n \sqrt{2 \mu_n C_{ox n} \frac{W}{L} I_0}$, $\frac{v_{g4}}{v_i} = \sqrt{\frac{\mu_n C_{ox n}}{\mu_p C_{ox p}}}$, $v_{g3} = v_i$

1.6:

A1.6

M_1 :- $I_0 = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}[V_{Dn} - V_T]^2$

M_2 :- $I_0 = \frac{1}{2}\mu_n C_{ox} N\frac{W}{L}[V_{Dx} - I_0 R - V_T]^2$

$\therefore V_{Dn} = V_T + \sqrt{\frac{2I_0}{\mu_n C_{ox} W/L}}$, $V_{Dx} = I_0 R + V_T + \sqrt{\frac{2I_0}{\mu_n C_{ox} N\frac{W}{L}}}$

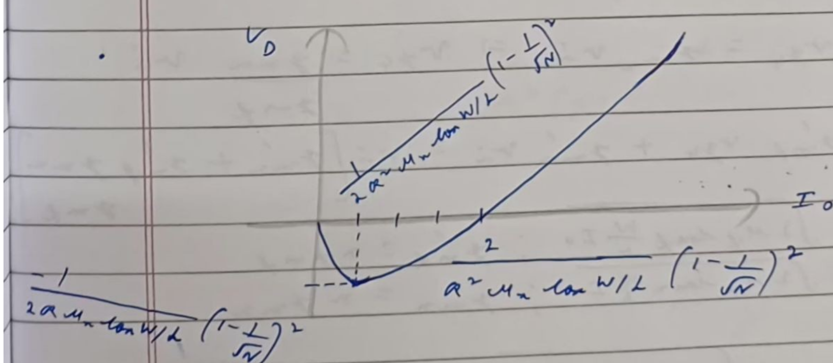
$$V_D = V_{Dx} - V_{Dn} = I_0 R - \sqrt{\frac{2I_0}{\mu_n C_{ox} W/L}}\left(1 - \frac{1}{\sqrt{N}}\right)$$

- $V_D = 0 \Rightarrow I_0 = \frac{2}{R^2 \times \mu_n C_{ox}\frac{W}{L}}\left(1 - \frac{1}{\sqrt{N}}\right)^2$

$$\therefore g_{m,M_1} = \sqrt{2I_0 \mu_n C_{ox} W/L} = \frac{2}{R}\left(1 - \frac{1}{\sqrt{N}}\right)$$

- $\frac{dV_D}{dI_0} = 0 \Rightarrow I_0 = \frac{1}{2R^2 \times \mu_n C_{ox} W/L}\left(1 - \frac{1}{\sqrt{N}}\right)^2$

$$\therefore V_{D,min} = -\frac{1}{2R \times \mu_n C_{ox} W/L}\left(1 - \frac{1}{\sqrt{N}}\right)^2$$

- 

1.7:

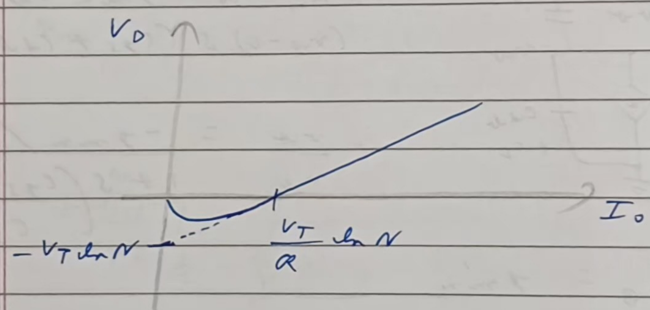
A1.7 $I_C = A J_S e^{V_{Dn}/V_T}$, $I_C = N A J_S e^{\frac{V_{DX} - I_0 R}{V_T}}$

$\therefore V_{Dn} = V_T \ln\left(\frac{I_C}{A J_S}\right)$, $V_{DX} = I_0 R + V_T \ln\left(\frac{I_C}{N A J_S}\right)$

$\therefore V_D = V_{DX} - V_{Dn} = I_0 R + V_T \ln\left(\frac{1}{N}\right)$

- $V_D = 0 \Rightarrow I_0 = \frac{V_T}{R} \ln N$

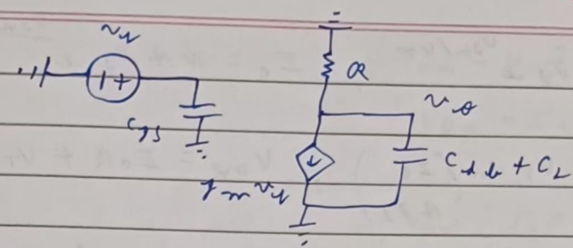
- $g_{m,Q_1} = \frac{I_C}{V_T} = \frac{\beta}{\beta+1} \frac{I_0}{V_1} = \frac{\beta}{\beta+1} \frac{\ln N}{R} \simeq \frac{\ln N}{R}$

- 

Graph is assumed to curve to (0, 0), because otherwise it would have 0 current for non-zero V_D.

1.8:

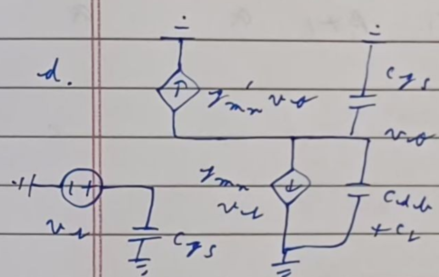
A 1.8 a) a.



$$v_o = -g_m v_i \times \left[R \,\|\, \frac{1}{s(C_{db}+C_L)} \right] = \frac{-g_m v_i R}{1 + s(C_{db}+C_L)R}$$

$$\therefore \omega_{-3dB} = \frac{1}{R(C_{db}+C_L)}$$

d.

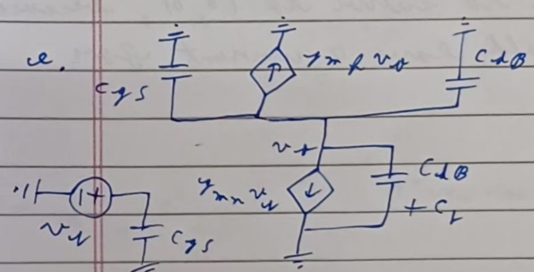


$$g'_{mn} v_o + g_{mn} v_i + (v_o - 0)\, s\,(C_{gs} + C_{db} + C_L) = 0$$

$$\therefore \frac{v_o}{v_i} = \frac{-g_{mn}/g'_{mn}}{1 + s\left(\dfrac{C_{gs} + C_{db} + C_L}{g'_{mn}}\right)}$$

$$\therefore \omega_{-3dB} = \frac{g'_{mn}}{C_{gs} + C_{db} + C_L}$$

e.

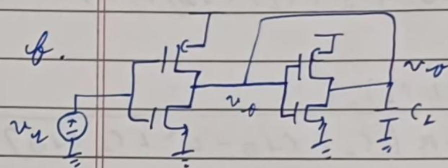


$$g_{mp} v_o + g_{mn} v_i + v_o \times s\,(C_{gs} + C_L + 2C_{dB}) = 0$$

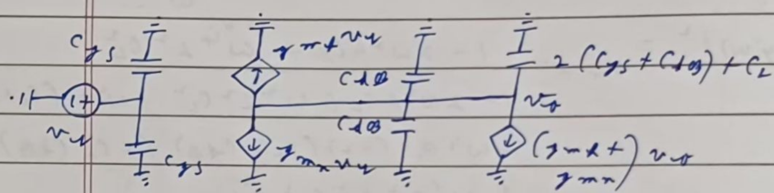
$$\frac{v_o}{v_i} = \frac{-g_{mn}/g_{mp}}{1 + s\left(\dfrac{C_{gs} + C_L + 2C_{dB}}{g_{mp}}\right)}$$

$$\omega_{-3dB} = g_{mp} / (C_{gs} + 2C_{dB} + C_L)$$

f.

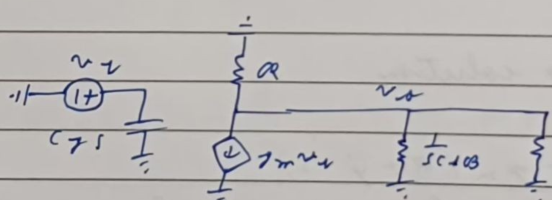


$$(g_{mn} + g_{mp})(v_o + v_i) + v_o \times s(2C_{gs} + 4C_{dB} + C_L) = 0$$



$$\frac{v_o}{v_i} = \frac{-1}{1 + s\left(\frac{2C_{gs} + 4C_{dB} + C_L}{g_{mp} + g_{mn}}\right)} \Rightarrow \omega_{-3dB} = \frac{g_{mn} + g_{mp}}{2C_{gs} + 4C_{dB} + C_L}$$

b) b.

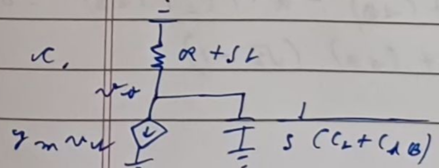


$$sL + \frac{1}{sC_L} = \frac{s^2LC_L + 1}{sC_L}$$

$$-g_m v_i = v_o\left(\frac{1}{R} + sC_{dB} + \frac{sC_L}{s^2LC_L + 1}\right)$$

$$\therefore \frac{v_o}{v_i} = \frac{-g_m R(s^2LC_L + 1)}{s^3LC_LC_{dB}R + s^2LC_L + sR(C_L + C_{dB}) + 1}$$

c.



$$-g_m v_i = v_o\left[s(C_L + C_{dB}) + \frac{1}{R + sL}\right]$$

$$\therefore \frac{v_o}{v_i} = \frac{-g_m(R + sL)}{s^2 L(C_L + C_{dB}) + s(C_L + C_{dB})R + 1}$$

1) b. $$H(j\omega) = \frac{-g_m R(1 - \omega^2 L C_L)}{1 - \omega^2 L C_L + j\omega R[C_L + C_{dB} - \omega^2 L C_L C_{dB}]}$$

$$\therefore |H(j\omega)|^2 = g_m^2 R^2 \frac{1 - 2\omega^2 L C_L + \omega^4 L^2 C_L^2}{1 - 2\omega^2 L C_L + \omega^4 L^2 C_L^2 + \omega^2 R^2 (C_L + C_{dB})^2 + \omega^4 R^2(-2)(C_L + C_{dB})(L C_L C_{dB}) + \omega^6 R^2 L^2 C_L^2 C_{dB}^2}$$

$d_1 = n_1$, $d_2 = n_2$ $\Rightarrow$ $R^2(C_L + C_{dB})^2 = 0$ and
$R^2(C_L + C_{dB}) \times L C_L C_{dB} = 0$

no solution

c. $$H(j\omega) = \frac{-g_m(R + j\omega L)}{1 - \omega^2 L(C_L + C_{dB}) + j\omega R(C_L + C_{dB})}$$

$$|H(j\omega)|^2 = g_m^2 R^2 \frac{(1 + \omega^2 L^2/R^2)}{1 + \omega^2[R^2(C_L + C_{dB})^2 - 2L(C_L + C_{dB})] + \omega^4 L^2 (C_L + C_{dB})^2}$$

$d_1 = n_1 \Rightarrow L^2 + 2LR^2(C_L + C_{dB}) - R^4(C_L + C_{dB})^2 = 0$

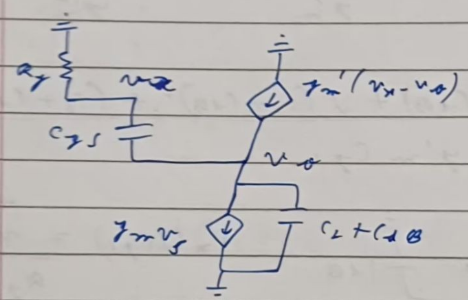
$\therefore L = R^2(C_L + C_{dB})(\sqrt{2} - 1)$

For ω_{3dB}:- $$|H(j\omega)|^2 = g_m^2 R^2 \frac{1+\omega^2 L^2/R^2}{1+\omega^2 L^2/R^2 + \omega^4 L^2 (C_L + C_{dB})^2} = \frac{g_m^2 R^2}{2}$$

$$\therefore \omega^4 L^2 (C_L + C_{dB})^2 - \omega^2 L^2/R^2 - 1 = 0$$

$$\therefore \omega_{3dB} = \frac{1}{R(C_L + C_{dB})}\sqrt{\frac{(3-2\sqrt{2}) \pm \sqrt{29-20\sqrt{2}}}{2(3-2\sqrt{2})}} = \frac{1.722}{R(C_L + C_{dB})}$$

d) d:



$$\frac{v_x}{R_g} = \frac{v_o}{R + 1/sC_{gs}}$$

$$\therefore v_x = \frac{s C_{gs} R_g v_o}{1 + s C_{gs} R_g}$$

$$\therefore g_m'(v_x - v_o) = \frac{v_o}{R_g + \frac{1}{sC_{gs}}} + g_m v_s + v_o s (C_L + C_{dB})$$

$$\therefore \frac{v_o}{v_s} = \frac{-g_m (1 + s C_{gs} R_g)}{g_m' + s[C_{gs} + C_L + C_{dB}] + s^2 C_{gs}(C_L + C_{dB}) R_g}$$

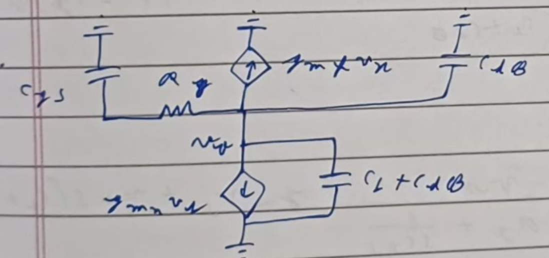
$$H(j\omega) = \frac{-g_m/g_m' \, (1 + j\omega R_g C_{gs})}{1 - \omega^2 \frac{C_{gs}(C_L + C_{dB}) R_g}{g_m'} + j\omega\left(\frac{C_L + C_{gs} + C_{dB}}{g_m'}\right)}$$

$$\therefore |H(j\omega)|^2 = \left(\frac{g_m}{g'_m}\right)^2 \times \frac{1 + \omega^2 R_g^2 C_{gs}^2}{1 + \omega^2\left[\left(\frac{C_{gs}+C_L+C_{dB}}{g_m'}\right)^2 - \frac{2C_{gs}(C_L+C_{dB})R_g}{g'_m}\right] + \omega^4 \frac{C_{gs}^2 (C_L+C_{dB})^2 R_g^2}{(g'_m)^2}}$$

$$c_1 = d_1 \Rightarrow R_g^2 C_{gs}^2 + 2R_g \frac{C_{gs}(C_L+C_{dB})}{g'_m} - \frac{(C_L+C_{dB}+C_{gs})^2}{g'^2_m} = 0$$

$$\therefore R_g = \frac{-(C_L+C_{dB}) + \sqrt{(C_L+C_{dB})^2 + (C_L+C_{dB}+C_{gs})^2}}{g'_m C_{gs}}$$

d.



$$v_x(sC_{gs}) = \frac{v_o}{R_g + \frac{1}{sC_{gs}}}$$

$$v_x = \frac{v_o}{1 + sC_{gs}R_g}$$

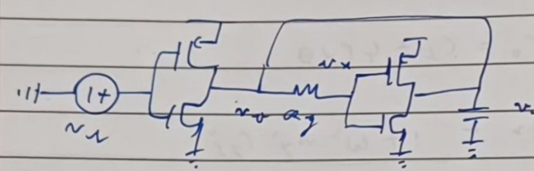
$$g_{mp}v_x + g_{mn}v_i + \frac{v_o}{R_g + \frac{1}{sC_{gs}}} + v_o s(C_L + 2C_{dB}) = 0$$

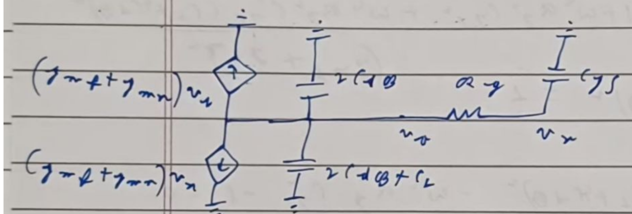
$$\therefore \frac{v_o}{v_i} = \frac{-g_{mn}(1+sC_{gs}R_g)}{g_{mp} + s(C_{gs}+C_L+2C_{dB}) + s^2 C_{gs}R_g(C_L+2C_{dB})}$$

Solving a similar set of equations for the maximally flat conditions as d. $(C_{dB} \to 2C_{dB})$,

$$\alpha_y = \frac{-(C_L+2C_{dB}) + \sqrt{(C_L+2C_{dB})^2 + (C_L+2C_{dB}+C_{gs})^2}}{g_{mp}\, C_{gs}}$$

f.





$$v_x (s C_{gs}) = \frac{v_o}{\alpha_y + \frac{1}{sC_{gs}}}$$

$$v_x = \frac{v_o}{1+\alpha_y C_{gs} s}$$

$$(g_{mp}+g_{mn})(v_i+v_x) + \frac{v_o}{\alpha_y+\frac{1}{sC_{gs}}} + v_o\, s(4C_{dB}+C_L) = 0$$

$$\therefore \frac{v_o}{v_i} = \frac{-(g_{mp}+g_{mn})(1+sC_{gs}\alpha_y)}{g_{mp}+g_{mn} + s(C_L+C_{gs}+4C_{dB}) + s^2\alpha_y C_{gs}(4C_{dB}+C_L)}$$

$$\therefore |H(j\omega)|^2 = \frac{1+\omega^2\alpha_y^2 C_{gs}^2}{1+\omega^2\left[\left(\frac{C_{gs}+C_L+4C_{dB}}{g_{mp}+g_{mn}}\right)^2 - \frac{2C_{gs}(C_L+4C_{dB})\alpha_y}{g_{mp}+g_{mn}}\right] + \omega^4\frac{C_{gs}^2(C_L+4C_{dB})^2\alpha_y^2}{(g_{mp}+g_{mn})^2}}$$

$$a_1 = d_1 \Rightarrow \alpha_y^2 C_{gs}^2 + \frac{2\alpha_y C_{gs}(C_L+4C_{dB})}{g_{mp}+g_{mn}} - \frac{(C_L+4C_{dB}+C_{gs})^2}{(g_{mp}+g_{mn})^2} = 0$$

$$\therefore \alpha_g = \frac{-(C_L + 4C_{dB}) + \sqrt{(C_L + 4C_{dB})^2 + (C_L + 4C_{dB} + C_{gs})^2}}{(g_{mp} + g_{mn})\, C_{gs}}$$

e) f. $C_i = C_{gs}$, $C_o = C_L + 4C_{dB}$

$$\therefore |H(j\omega)|^2 = \frac{1 + \omega^2 \alpha_g^2 C_{gs}^2}{1 + \omega^2 \alpha_g^2 C_{gs}^2 + \dfrac{\omega^4 \alpha_g^2 C_{gs}^2 (C_L + 4C_{dB})^2}{(g_{mp} + g_{mn})^2}}$$

$\omega_{-3dB} \Rightarrow |H(j\omega)|^2 = \frac{1}{2}$

$$\therefore \omega^4 \frac{\alpha_g^2 C_{gs}^2 (C_L + 4C_{dB})^2}{(g_{mp} + g_{mn})^2} - \omega^2 \alpha_g^2 C_{gs}^2 - 1 = 0$$

$$\frac{\omega^4 \alpha_g^2 C_i^2 C_o^2}{(g_{mp} + g_{mn})^2} - \omega^2 \alpha_g^2 C_i^2 - 1 = 0$$

where $\alpha_g = \dfrac{-C_o + \sqrt{C_o^2 + (C_o + C_i)^2}}{(g_{mp} + g_{mn})\, C_i}$

Solving, $\omega^2 = \dfrac{(g_{mp} + g_{mn})^2}{2C_o^2}\left[1 + \dfrac{\sqrt{6C_o^2 + (C_o + C_i)^2 - 2C_o\sqrt{C_o^2 + (C_o + C_i)^2}}}{\sqrt{C_o^2 + (C_o + C_i)^2} - C_o}\right.$

↳ $C_o = C_i = \frac{C}{2} \Rightarrow \omega^2 = 7.688 \left(\dfrac{g_{mp} + g_{mn}}{C}\right)^2$

$\therefore \omega_{-3dB} = 2.772\, (g_{mp} + g_{mn})/C$

↳ $C_o = \frac{3}{4}C$, $C_i = \frac{C}{4} \Rightarrow \omega^2 = 3.699\, (g_{mp} + g_{mn})^2 / C^2$

$\therefore \omega_{-3dB} = 1.92\, (g_{mp} + g_{mn})/C$